

Silvia Sifath

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Research Interest

Artificial Intelligence, Deep Learning, Machine Learning, Agent-based Modelling, Bioinformatics, Fuzzy Expert System, Natural Language Processing

Academic Credentials

- **M.Sc.** in Computer Science and Engineering (June 2023-May 2026) **CGPA – 3.96** (1st)
Institute – University of Barishal
- **B.Sc.** in Computer Science and Engineering (Jan 2017 – March 2023) **CGPA - 3.88** (1st)
Institute – University of Barishal
- **HSC** in Science group (2014-2016) **GPA – 5.00** (Scholarship)
Institute – Guthia Ideal College
- **SSC** in Science group (2012-2014) **GPA – 5.00** (Scholarship)
Institute – Banaripara Secondary Girls School

Working Experience

Lecturer: February 01, 2025 to continue

Organization: Trust Univeristy, Barishal

Assistant Proctor: April 01, 2025 to continue

Organization: Trust Univeristy, Barishal

Research Fellow: Masters Thesis

Organization: ICT Division (Under University of Barishal, Barishal)

Supervised by: Assoc. Prof. Dr. Md Manjur Ahmed

Research Assistant: January'01, 2022 to January'31, 2025

Organization: University of Barishal, Barishal, Bangladesh

Supervised by: Assoc. Prof. Dr. Md Manjur Ahmed

Research Assistant: July'01, 2021 to December'31, 2021

Organization: University of Barishal, Barishal, Bangladesh

Supervised by: Assoc. Prof. Dr. Rahat Hossain Faisal

Research Assistant: March'01, 2021 to May'31, 2021

Organization: University of Barishal, Barishal, Bangladesh

Supervised by: Assoc. Prof. Dr. Md Manjur Ahmed

Publication

Adaptive Similarity-Based Non-negative Matrix Factorization with Normalization for Image Clustering Accepted in Springer (International Journal of Machine Learning and Cybernetics, Volume 17, April 2026, 285) Doi: <https://doi.org/10.1007/s13042-026-03119-z>

Automatic classification of orange fruit diseases using deep neural network model Published in Elsevier (Journal of Agriculture and Food Research, Volume 27, May 2026, 102712) Doi: <https://doi.org/10.1016/j.jafr.2026.102712>

Recurrent Neural Network-based Multiclass Cyberbullying Classification Published in Elsevier (Natural Language Processing Journal, Volume 9, December 2024, 100111) Doi: <https://doi.org/10.1016/j.nlp.2024.100111>

A Robust Ensemble Framework for Alzheimer's Disease Prediction Using Feature Engineering and Optimized Voting Classifier Accepted as lecture notes in Scopus-indexed proceedings published by Springer at the ICETCS and ITSS-IoE conferences, 2025, held at Science Park, Wolverhampton, UK. Author: Silvia Sifath, Md Arafatur Rahman, Mohammed Abdur Rahman, Md.Manjur Ahmed*, and Selwa Al-Hazzaa

Intelligent Learning Behavior Profiling System for Student Based on Fuzzy Agent Model Published as a book chapter in Data-Driven Applications for Emerging Technologies, a book series by Taylor & Francis. Doi: [10.1201/9781003605508-2](https://doi.org/10.1201/9781003605508-2)

Student learning behavior profiling system based on fuzzy agent-based system Submitted in Springer(International Journal of Artificial Intelligence in Education) on 04 February, 2026. Author: Silvia Sifath, *Md Manjur Ahmed, Md Rahat Hossian Faisal

Regional Language Classification in Bangladesh Using NLP and Machine Learning Techniques Submitted in Elsevier (Arabian Journal for Science and Engineering) on November 2025. Author: Silvia Sifath, **Sajeeb Saha, *Md. Rashid Al Asif, Sohely Jahan, Md. Samsuddoha, Tania Islam

A Synergistic Framework Combining Bloom's Taxonomy and Explainable AI for Question Classification Using Machine Learning Submitted in Springer (International Journal of Artificial Intelligence in Education) on September 2025. Author: Silvia Sifath*, Jahangir Alam

A Modular Fuzzy Expert System for Chemotherapy Drug Dose Scheduling Published in Elsevier (Healthcare Analytics, Volume 3, November 2023, 100139). Doi: <https://doi.org/10.1016/j.health.2023.100139>

Natural Convective Heat and Mass Transfer Flow in a Doubly Stratified High Porosity Medium Published in Willy (Engineering Reports, Volume 8(4), April 2026, p.e70742.). Doi: [10.1002/eng2.70742](https://doi.org/10.1002/eng2.70742)

Multi-objective Optimization and Feedback-based Chemotherapy Drug Scheduling Published in IEEE (2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), pp.1 -6, June 24, 2024). Doi: [10.1109/ICAEEE62219.2024.10561667](https://doi.org/10.1109/ICAEEE62219.2024.10561667)

Affordable Early Detection of Autism Spectrum Disorder Using Innovative Eye-Tracking Methods Published in IEEE Xplore(27th International Conference on Computer and Information Technology (ICCIT)) on 11 June, 2025. Doi: <https://doi.org/10.1109/ICCIT64611.2024>

Enhanced Fuzzy Expert Control for Cancer Chemotherapy Drug Dose Scheduling Published in Barishal Univ. J. Bio-Sci. 3(1): 47-64, 2024 (June). Doi: [10.13140/RG.2.2.12355.82721](https://doi.org/10.13140/RG.2.2.12355.82721)

Optimization of Chemotherapy Drug Regimens with Genetic Algorithm and Dose Optimizer Published in Barishal Univ. J. Bio-Sci. 3(1): 65-74, 2024 (June). Doi: [10.13140/RG.2.2.12146.11208](https://doi.org/10.13140/RG.2.2.12146.11208)

Activities and Achievements

- ICT Fellowship (For Masters Thesis) | 2024
- Dean's Award (For Academic Excellence in the Science and Engineering Faculty) | October 2023
- Participating Member | Developed PregnaCare; selected from 303 teams at the Harvard Hackathon and presented the project | April 2025
- Certificate of climate and human rights (online) | Amnesty international - April 2023
- EDGE Project Participant | ICT Division, Government of Bangladesh | Aug 2024 – Nov 2024
- Board scholarship for HSC (Based on public examination results in the Science Faculty) - 2016
- Board scholarship for SSC (Based on public examination results in the Science Faculty) - 2014
- JSC scholarship (For extraordinary result in public examination) – 2012

Project

Brain Tumor Prediction using Machine Learning and Deep learning (Jan 2022 - Mar 2022)

- Analyzed the dataset to find out the correlation and further deciding the most suitable ML and deep learning algorithm to apply.

SuperHut (March 2019 – June 2019)

- Developed an online shopping application with .net framework using C2C model concept

Scientific Introduction (Dec 2018 - Jan 2019)

- Developed an application to find out the scientific introduction of each organism.
- Developed Java Swing application and MySQL database.

Technical Skills

Languages: Java, Python, C, C++, MATLAB, HTML, SQL, PHP

Frameworks/ORM: Django(learning), Java Swing, .net, Hadoop(basic)

Tools: Eclipse, MySQL, Latex, MATLAB, PyCharm, Visual studio, Oracle Database, Power Bi, Tableau public

Software Engineering: SDLC, SRS, System design, OOP, Design Patterns, Requirement Engineering, Problem Solving

Standardized Test Scores

International English Language Testing System (IELTS) - 10 June, 2023

Overall	Listening	Reading	Writing	Speaking
6.5	6.0	7.5	6.0	6.0

Personal Details

Father's Name	: Md Nurul Islam	Mother's Name	: Aleya Begum
Date of Birth	: 12/12/1998	Gender	: Female
Marital Status	: Single	Nationality	: Bangladeshi
National Id No	: 7354147444	Religion	: Islam
Permanent address	: Mojidbari, Ward No-6, Banaripara, Mojidbari-8530, Barishal		
Current Location	: Holding NO:233, Gosher Bari, Ward NO:04, Banaripara 8530, Banaripara, Barishal		

References

Dr. Md. Manjur Ahmed

Associate Professor,
Department of Computer Science & Engineering,
University of Barishal
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Phone: 01851924944

Dr. Rahat Hossain Faisal

Chairman & Associate Professor,
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